

TIM 147 Project Phase II Predictive Analytics Report

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1 Executive Summary

The Client Health Analytics Initiative focuses on predicting churn, understanding what drives lifetime value, and identifying patterns across client segments. After Phase I clarified what data is available and usable, this phase focuses on building models that support Alan’s need for clearer lifetime value estimates and Nate’s need for a structured way to prioritize outreach before accounts are lost.

For lifetime value, we defined LTV as cumulative realized invoice revenue rather than projected contract totals. This keeps the estimate tied to actual revenue collected and avoids overstating accounts that churn early. We compared Linear Regression, Ridge Regression, and Random Forest models. Random Forest performed best, reducing average error to about \$26.9K and explaining roughly 97% of the variation in LTV. Strategic accounts average around \$646K in lifetime value, Core around \$383K, Growth around \$263K, and SMB around \$186K. Strategic clients are worth more than three times as much as SMB clients over the full relationship. The strongest drivers of LTV are total contract value and recent utilization, showing that larger contracts and higher product usage are closely tied to long-term revenue.

For churn, we modeled 558 accounts with an overall churn rate of 22.9%. The risk varies sharply by segment. SMB churn is about 39%, while Strategic churn is about 14%. Logistic Regression and Random Forest produced nearly identical results, with about 97% accuracy and 96% recall on a held-out test set. Because performance was effectively the same, Logistic Regression was selected for its interpretability. The strongest predictor of churn is days since last Client Success activity. Accounts that have not been contacted recently, have fewer active contracts, experience slower support resolution, and show lower usage and invoice activity are much more likely to churn. All 430 active accounts were scored and ranked, producing a prioritized list that Nate’s team can use to guide outreach.

These results suggest clear next steps. Strategic and Core accounts generate the majority of long-term revenue and should receive consistent engagement. Engagement signals such as inactivity and declining usage can serve as early warnings instead of waiting for visible problems. Improving support response times might also help reduce churn risk. The SMB segment has both the lowest (approximately \$186K on average) and the highest churn rate (approximately 39%), which suggests the company should carefully evaluate how much acquisition and retention effort is focused on this group.

There are important limitations. The churn label tells us whether an account has ever churned in the past. It does not predict whether an account will churn within a specific future time frame such as the next 60 or 90 days. Several planned features, including NPS and detailed engagement metrics, were excluded due to missing data. The models highlight strong correlations but do not prove causation. Even with these constraints, the analysis provides clear, data-backed guidance to support pricing, sales targeting, and client success decisions.

2 Lifetime Value Model

2.1 Target Variable

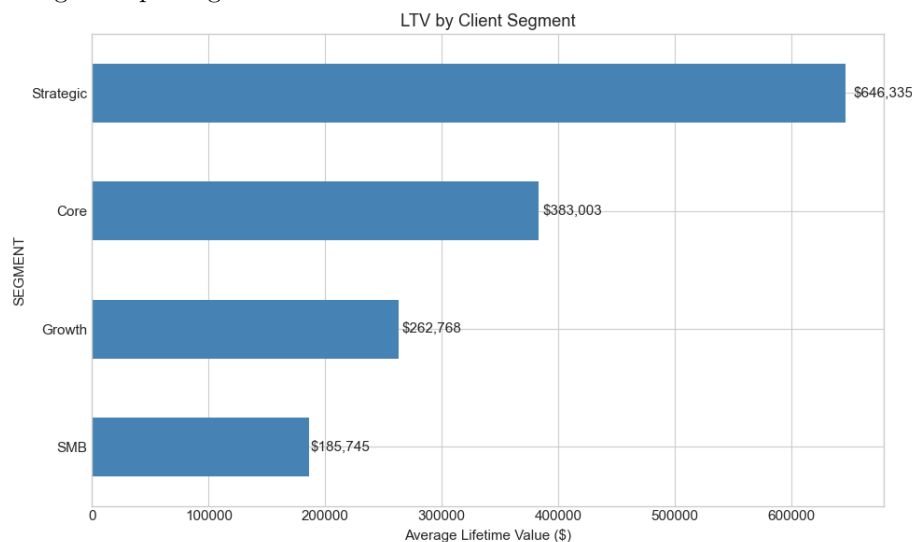
While looking at the data provided, we saw two main contenders for a target value for modeling Lifetime Value of an account: `TOTAL_VALUE` or `INVOICES`. While `TOTAL_VALUE` provides expected revenue, there is a possibility that clients could churn early making `TOTAL_VALUE` not fully representative of an account.

Because of this, we decided a better approach would be to take the accumulation of what has been paid (or is being collected) by aggregating the values for each account of `INVOICES`.

By focusing on the realized value of an account, we could compare the value of the account against any previous costs to acquire the client and for any services.

2.2 Feature and Aggregation

Before we constructed a predictive model, we wanted to profile the different segments and see what was their average lifetime value (LTV). We found the following LTV per segment:



When adding in the number of accounts, we get the following summary:

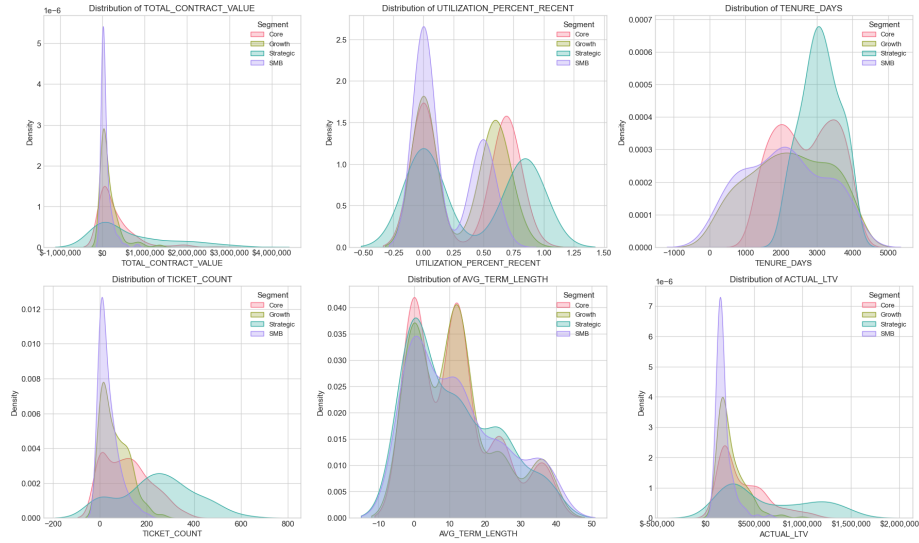
Segment	Accounts	LTV
SMB	87	\$185,745
Growth	177	\$262,768
Core	222	\$383,003
Strategic	72	\$646,335

We see the strategic segment had the highest LTV, more than triple that of the SMB segment despite the fact that Strategic has the fewest accounts. To fully see what led to the values and capture the costs it takes to support and retain a client across different segments, we aggregated data from 8 warehouse tables into 29 primary features which were categorized into 4 sections:

1. Usage: e.g. UTILIZATION and TREND
2. Commitment: e.g. TENURE_DAYS and AVG_TERM_LENGTH
3. Engagement: e.g. TICKET_COUNT and TOTAL_RESOLUTION_TIME
4. Account Stats: e.g. EMPLOYEE_COUNT and ANNUAL_REVENUE

In order to make sure these features were actually relevant across the different segments, we mapped their distribution of the different features:

Predictive Modeling: Feature Distributions Across Client Segments



This visualization confirmed that the Strategic and Core segments followed different distributions than the SMB segment, supporting our future use of the Random Forest Model.

2.3 Models

We used three different models with the chosen feature matrix and target variable: Linear Regression, Ridge Regression, Random Forest. This gave us the following values:

— Linear Regression Results —
 Average Error (MAE): \$57,959.88
 Model Variance Explained (R2): 89.16%

— Ridge Regression Results —
 Average Error (MAE): \$57,747.46
 Model Variance Explained (R2): 89.19%

— Random Forest Results —
 Average Error (MAE): \$26,850.16
 Model Variance Explained (R2): 97.51%

We had initially started with a linear regression in order to have a baseline for interpretability. The second main model was a Random Forest model. This allowed us to better see how the different features interacted with different segments. Unlike the linear models that assumed a fixed relationship for all variables, the Random Forest model allowed us to identify the decision branches. For example, `TICKET_COUNT` could signal for healthy integration in a Strategic account, but it could be a signal for friction in an SMB account.

While tuning this model, we optimized it for 200 trees with no max depth and a minimal sample split of 2. This allowed us to be able to better split the segments.

Core MAE: \$14,745.12
 Growth MAE: \$12,370.86
 Strategic MAE: \$37,479.99
 SMB MAE: \$8,818.64

We can see that the most accurate of the three models is the Random Forest Model. Even in the Strategic segment, where the error is highest in dollar terms, it represents less than 6% of the segment's LTV.

2.4 Evaluation

When analyzing the Lifetime Value Estimation (LTV) for the different accounts withing different segments, we see an interesting "Upsell Gap." By comparing the predicted LTV against the realized revenue, we identified accounts that are currently under-monetized relative to their usage patterns.

--- SEGMENT REPORT ---					
	SEGMENT	Avg_Actual_LTV	Avg_Predicted_LTV	Avg_Upsell_Gap	Client_Count
2	SMB	\$185,744.57	\$188,436.57	\$2,692.00	107
0	Core	\$383,002.79	\$383,325.58	\$322.80	290
1	Growth	\$262,767.58	\$262,844.79	\$77.21	207
3	Strategic	\$646,334.93	\$637,545.77	\$-8,789.17	87

We found that the primary drivers of LTV are:

TOTAL_VALUE: 0.88 correlation
RECENT_UTILIZATION: 0.72 correlation
TICKET_COUNT: 0.62 correlation

The high correlation between TICKET_COUNT and LTV suggests that high-support accounts are not necessarily less-valuable. Instead, it proves that for Vantage, high engagement with support teams is a leading indicator for deep product integration with long-term financial commitment.

2.5 Suggestions

Based on the model's output and the identified Upsell Gaps, we offer the following recommendations:

1. **Target the SMB Value:** The SMB segment shows the highest positive Upsell Gap (avg. +\$2,692 per account). We recommend the Sales team target these 107 accounts for tier upgrades since their resource consumption surpasses their current contract value.
2. **Retention for Strategic Accounts:** Strategic accounts show a negative gap (-\$8,789), meaning they are currently maximizing (or slightly over-utilizing) their contract value. Given their high total LTV, we should ensure renewal, rather than pursuing aggressive upselling.
3. **Usage-Based Triggers:** Since RECENT_UTILIZATION is the second strongest driver of LTV, we suggest implementing automated alerts when an account's usage exceeds 80% of their tier limit, as this is the most reliable predictor of a future LTV jump.

3 Churn Model

3.1 Goal

For Phase II we built a classification model to help Nate's Client Success team figure out which active accounts are most likely to churn. The team's current process is mostly reactive. They respond to warning signs once they're already visible, which means some accounts probably slip through. The goal was to give them a ranked list they can actually use to get ahead of that, so outreach is happening before an account has already checked out.

3.2 Churn Label

We used the IS_CHURNED flag and STATUS field from the ACCOUNTS table to define the target. Accounts with STATUS = 'churned' were labeled 1,

accounts with STATUS = ‘active’ were labeled 0. We dropped prospects, implementation_only, and lost accounts because those never really went through the full customer lifecycle, so including them felt like it would confuse the model more than help it.

After that filtering we had 558 accounts total. 430 active, 128 churned. Overall churn rate is 22.9%. The one caveat here is that this is an all-time label (it captures whether an account ever churned, not whether it churned in the last quarter or the next 90 days). That’s a real limitation and something we’d want to address in a future version. For now we tried to build in time-sensitivity through the features, things like recency of last CS activity and whether engagement has been going up or down lately.

Churn rate by segment:

Segment	Accounts	Churn Rate
SMB	87	39.1%
Growth	177	23.7%
Core	222	18.9%
Strategic	72	13.9%

SMB is the obvious outlier here, with nearly a 40% churn rate compared to 14% for Strategic. Smaller clients probably have less internal advocacy for the product and fewer contractual reasons to stay.

3.3 Features and Aggregation

Features came from 8 warehouse tables, all aggregated to one row per account. We wanted a mix of static attributes and behavioral signals since churn tends to show up in behavior before it shows up anywhere else.

- **Account profile:** segment, employee count, annual revenue, account age
- **Contracts:** contract count, ACV totals and averages, active vs. cancelled counts
- **Support tickets:** volume, resolution time, satisfaction score, open ticket count
- **Service utilization:** usage levels, hours used, trend slope, recent vs. historical ratio
- **Engagement scores:** overall score, trend, recent vs. historical ratio
- **CS activities:** activity and meeting counts, days since last touchpoint
- **Invoices:** invoice count, overdue count, total invoiced

We had also planned to use NPS scores, several engagement sub-scores (login frequency, feature adoption, support engagement, training completion), and auto-renew percentage. All of those got dropped in preprocessing because the missing value rates were too high to impute reliably. So the model ended up running on 29 features rather than the full set we designed.

3.4 Models

3.4.1 Logistic Regression

Logistic regression was our first model. The main appeal is interpretability. The coefficients tell you directly which features drive churn and in which direction. We tuned the regularization parameter C from 0.001 to 100 and the best value was $C = 1$.

3.4.2 Random Forest

Random forest was the second. It works by building many decision trees on random data subsets and aggregating their outputs, so it can catch non-linear effects and interactions between features. Tuning covered `n_estimators` (100, 200), `max_depth` (5, 10, 15, None), and `min_samples_leaf` (2, 5, 10). Best setup: 200 trees, `max_depth` = 5, `min_samples_leaf` = 10.

We ran both mainly as a cross-check. When two very different algorithms produce similar results, it's harder to write off as a modeling artifact. In this case they agreed almost exactly on both performance metrics and feature importance, so that was useful confirmation.

3.5 Results

Tested on a held-out 20% split (112 accounts):

Model	Accuracy	Precision	Recall	F1	AUC
Logistic Regression	97.3%	92.6%	96.2%	94.3%	0.990
Random Forest	97.3%	92.6%	96.2%	94.3%	0.998

Logistic regression made 3 total errors on 112 accounts (2 false positives, 1 false negative). Both models were essentially tied so we went with logistic regression. Same performance, more explainable.

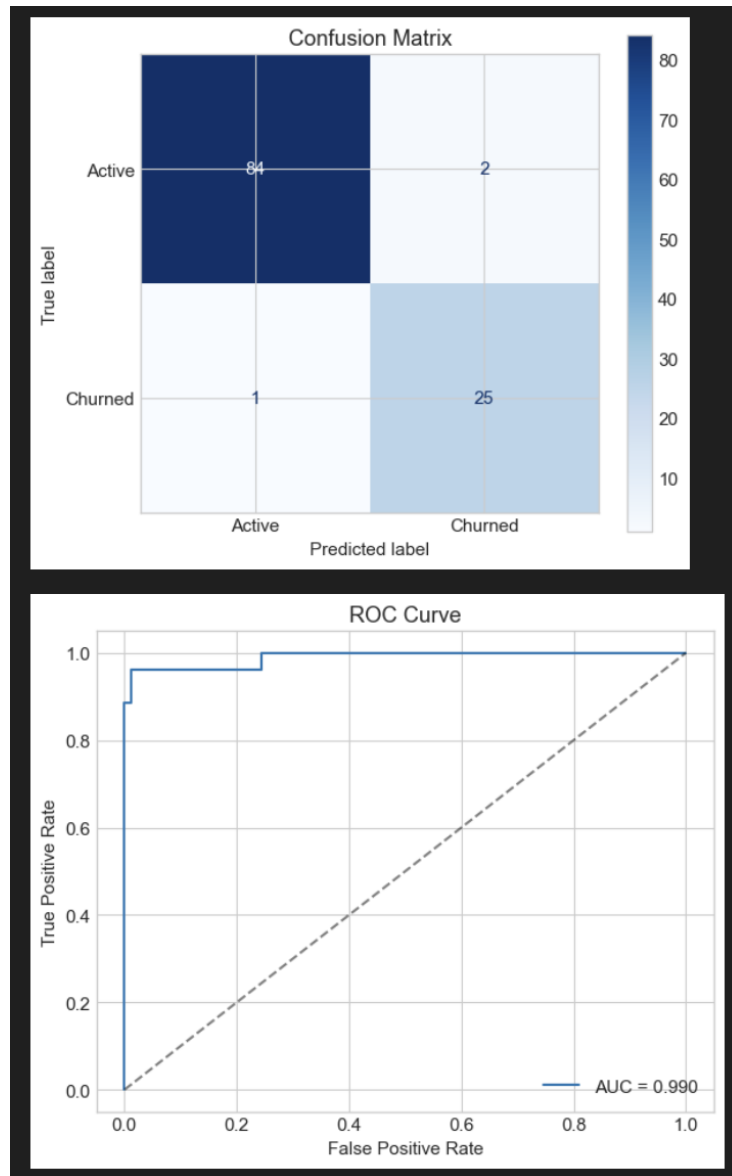


Figure 1: Logistic Regression: Confusion Matrix and ROC Curve (AUC = 0.990)

3.6 Key Drivers

The correlation analysis and model coefficients were consistent with each other, which gave us more confidence in the findings. Top three predictors:

- **Days since last CS activity:** strongest signal by a good margin. Accounts that haven't been contacted recently churn at much higher rates.
- **Active contract count:** fewer active contracts correlates strongly with churn. Could mean the relationship is already winding down before the official departure.
- **Average support resolution hours:** longer resolution times go with higher churn risk. Slow support seems to be a real driver, not just a coincidence.

Lower invoice counts and lower product usage hours also showed up in both models. Those are probably picking up on disengagement more broadly. An account that isn't using the product much and isn't being billed much isn't really an active customer in any meaningful sense.

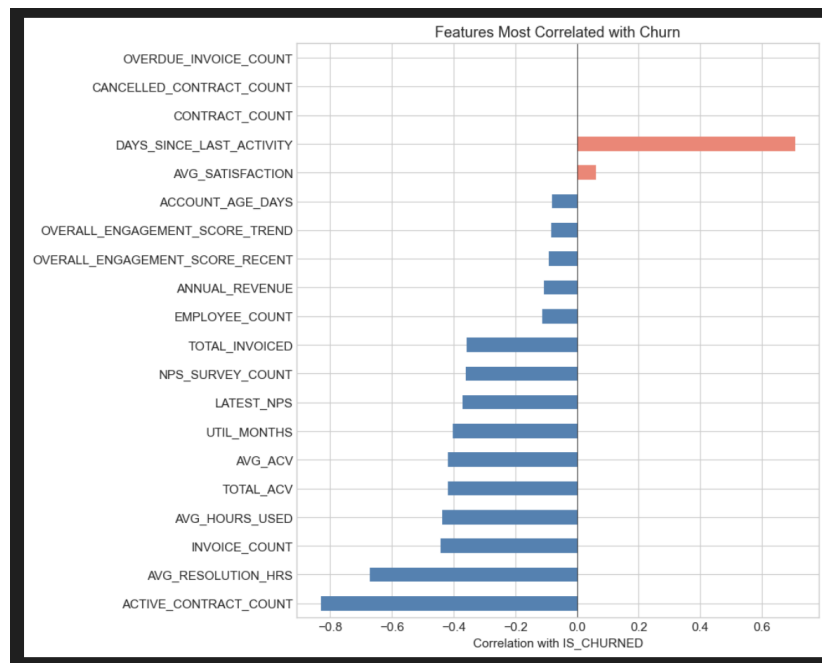


Figure 2: Features Most Correlated with Churn

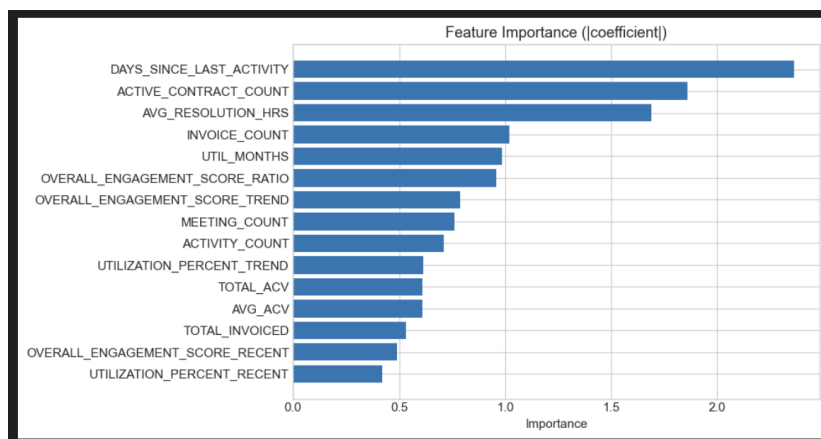


Figure 3: Logistic Regression Feature Importance (—coefficient—)

3.7 Risk Scoring and Outreach List

All 430 active accounts were scored and bucketed into tiers: 1 High-risk, 8 Medium-risk, 421 Low-risk. The full list is in `churn_risk_list.csv`.

The 9 High and Medium accounts are the most pressing. For threshold decisions, raising the cutoff means fewer accounts to contact but higher confidence in each flag, and lowering it gives broader coverage at the cost of more volume. That tradeoff is worth discussing with Nate depending on how much outreach bandwidth the team actually has.

3.8 Limitations

- The churn label is all-time, not tied to a future window. A production version should predict churn within a specific horizon like 60 or 90 days.
- Several planned features had to be dropped due to missing data (NPS, engagement sub-scores, auto-renew %). Better data coverage in the warehouse would likely improve model quality.
- These are correlations, not causal relationships. More CS touchpoints might reduce churn, but we can't prove that from this model alone.
- Adding product telemetry, renewal pipeline data, and competitive loss reasons would probably strengthen early warning capability going forward.